

Cancer Hotspot Mutation Scan – Neoantigen Epitope Prediction

Model: Deep FFN (MHCflurry-trained, 91.9% accuracy, 89.6% CV)

Allele: HLA-A*02:01

Scope: 16 hotspot mutations across p53 (7) and KRAS (9)

Date: June 15, 2026

Summary

Category	Count	Clinical Impact
Neopeptides CREATED	2	New immunotherapeutic targets
Epitopes ENHANCED	2	Stronger existing binders
Epitopes DESTROYED	3	Immune evasion mechanism
No epitope change	9	Silent for MHC-I binding
Total	16	

Neopeptides Created (NB -> SB/WB)

These mutations create entirely new HLA-A*02:01 epitopes – candidates for personalized cancer vaccines.

1. p53 R248W – CREATED neopeptide

Property	Value
Mutation	R > W at position 248
Clinical context	Most frequent p53 mutation across all cancers
WT peptide	MNRRPILTI (NB, score 0.002)
Mutant peptide	MNWRPILTI (WB, score 0.409)
Effect	Non-binder -> Weak Binder (+0.407)
p2 anchor	N (weak)
p9 anchor	I (canonical)
Mechanism	R>W at p3 introduces bulky hydrophobic residue, improving MHC groove fit

2. KRAS G12V – CREATED neoepitope

Property	Value
Mutation	G > V at position 12
Clinical context	Pancreatic and lung adenocarcinoma driver
WT peptide	YKLVVVGAG (NB, score 0.000)
Mutant peptide	YKLVVVGAV (WB, score 0.475)
Effect	Non-binder -> Weak Binder (+0.475)
p2 anchor	K (non-canonical)
p9 anchor	G -> V (becomes canonical!)
Mechanism	G>V at p9 replaces glycine with valine – perfect C-terminal anchor

Epitopes Enhanced (WB -> SB)

These mutations substantially boost existing weak binders to strong binders.

3. KRAS G12V – ENHANCED epitope

Property	Value
WT peptide	LVVVGAGGV (WB, score 0.509)
Mutant peptide	LVVVGAVGV (SB, score 0.816)
Effect	WB -> SB (+0.307)
p9 anchor	G -> V (canonical anchor created)

4. KRAS G12C – ENHANCED epitope

Property	Value
Mutation	G > C at position 12
Clinical context	Lung adenocarcinoma – druggable (sotorasib, adagrasib)
WT peptide	LVVVGAGGV (WB, score 0.509)

Property	Value
Mutant peptide	LVVVGACGV (SB, score 0.816)
Effect	WB -> SB (+0.307)
Clinical implication	Enhanced SB may synergize with KRAS G12C inhibitors for combination immunotherapy

Epitopes Destroyed (SB/WB -> NB)

These mutations eliminate existing epitopes – potential immune evasion mechanisms.

5. KRAS G13D – DESTROYED epitope

Property	Value
Mutation	G > D at position 13
Clinical context	Colon cancer
WT peptide	GVGKSALTI (WB, score 0.471)
Mutant peptide	DVDKSALTI (NB, score 0.001)
Effect	WB -> NB (-0.470)
Mechanism	G>D at p2 introduces negatively charged aspartate – destroys p2 anchor

6. KRAS A146T – DESTROYED epitope

Property	Value
Mutation	A > T at position 146
Clinical context	Colon cancer
WT peptide	GIPFIETSA (WB, score 0.436)
Mutant peptide	GIPFIETST (NB, score 0.020)
Effect	WB -> NB (-0.416)
Mechanism	A>T at p9 – threonine is a weaker anchor than alanine

7. p53 R249S – DESTROYED epitope

Property	Value
Mutation	R > S at position 249
Clinical context	Aflatoxin-associated hepatocellular carcinoma
WT peptide	GMNRRPILT (WB, score 0.401)
Mutant peptide	GMNRSPILT (NB, score 0.245)
Effect	WB -> NB (-0.156)
Mechanism	R>S at p5 – polar to small polar change affects TCR-facing residue

Silent Mutations (No Epitope Change)

Mutation	Reason
p53 R175H	Mutation outside 9-mer windows with binding potential
p53 G245S	Mutation in non-anchor position of non-binding windows
p53 R273H	DNA contact mutant – no A*02:01 epitope overlap
p53 R282W	LFS hotspot – no A*02:01 epitope overlap
p53 Y220C	Structural cavity mutant – no epitope overlap
KRAS G12D	G>D at p9 destroys anchor (like G13D but different window)
KRAS G12R	G>R at p9 introduces charged residue – destroys anchor
KRAS Q61H/L/R	Q61 mutations in non-binding windows

Clinical Recommendations

Tier 1 – Validate Immediately

Rank	Neopeptide	Mutation	Rationale
1	MNWRPILTI	p53 R248W	Most common p53 mutant; pan-cancer target
2	LVVVGAVGV	KRAS G12V	Pancreatic/lung cancer; enhanced to SB
3	LVVVGACGV	KRAS G12C	Druggable; combination therapy potential

Tier 2 – Mechanistic Investigation

Rank	Finding	Mutation	Question
4	GVGKSALTI destroyed	KRAS G13D	Does epitope loss contribute to immune evasion in MSI-H colon cancer?
5	GIPFIETSA destroyed	KRAS A146T	Same question for A146T colon cancers

Validation Protocol

1. **In silico:** Confirm with netMHCpan-4.1 and MHCflurry consensus
2. **In vitro:** T2 binding assay (HLA-A*02:01 stabilization)
3. **Ex vivo:** IFN-gamma ELISpot with patient PBMCs
4. **In vivo:** HLA-A*02:01 transgenic mouse immunization

Methods

- **Model:** Deep FFN (360-180-90-45-3), MHCflurry 2.2.0 trained, 91.9% accuracy
- **Scan method:** 9 sliding 9-mer windows spanning each mutation position
- **Comparison:** WT vs mutant binding score ($SB1.0 + WB0.5$)
- **Significance threshold:** delta score > 0.1 AND class change required

Output Files

File	Contents
data/mutation_scan_results.csv	Full 144-row results with all windows
mutation_scan_report.md	This report

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